Name_____ Date_____ Period_____

LESSON 5-1 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Area of Parallelograms

Maryland Standard 6.G.A.1 · Learning target – I can find the area of a parallelogram by multiplying the base by the perpendicular height.

REMEMBER

- Cut a triangle off a parallelogram and slide it over to make a rectangle: **A = b × h**.
- The height *h* is **perpendicular** to the base. Do not use the slanted side.
- Area is in **square units** (cm², in², ft²).

WORKED EXAMPLE

Find the area: base 15 in, slanted side 8 in, height 6 in.

Step 1 Use the perpendicular height: h = 6 (skip the 8 in side)

Step 2 A = b × h = 15 × 6

Step 3 Multiply: 15 × 6 = 90

Answer **90 in²**

WORD BANK

base – the side you measure from

height – straight up from the base, at a square corner

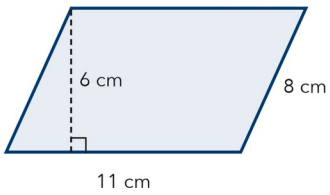
slanted side – the tilted side; not the height

area – space inside, in square units

PRACTICE

show your work · label units

1. Find the area of the parallelogram.



Step 1 Find the base and the height (with the square corner).

Step 2 Multiply base × height. Do not use 8 cm.

Show your work.

Answer_____ cm²

2. Find the area of each parallelogram with base *b* and height *h*.

Step 1 Multiply b × h for a, b, and c.

Step 2 For d, divide the area by the height.

a. b = 12 ft, h = 7 ft _____ ft²

b. b = 9 m, h = 9 m _____ m²

c. b = 20 in, h = 4 in _____ in²

d. A = 63 m², h = 7 m; b = _____ m

3. Parking Lot. A parking lot has 10 spaces. Each space is a parallelogram: base 9 ft, height 18 ft. What **total** area do the 10 spaces cover?

Step 1 Find the area of one space.

Step 2 Multiply by 10.

Show your work.

Answer_____ ft²

4. Mural. A mural is painted on a parallelogram-shaped wall section. The area is 108 ft² and the base is 12 ft. What is the perpendicular height?

Step 1 Area = base × height, so 108 = 12 × ___.

Step 2 Divide 108 by 12.

Answer_____ ft

5. Find the Error. A parallelogram has a base of 10 cm, a slanted side of 7 cm, and a height of 5 cm. Kiara says its area is 70 cm². Explain her mistake and find the correct area.

Frame *Kiara used the ___ side instead of the ___. The area is ___ × ___ = ___ cm².*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Find a parallelogram shape at home or on the street (a tile, a sign, a parking space). Show someone where its base and height are.

Name_____ Date_____ Period_____

LESSON 5-1 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Area of Rhombuses

Maryland Standard 6.G.A.1 · Learning target – I can find the area of a rhombus using the base and height or the two diagonals.

REMEMBER

- A rhombus is a parallelogram, so **A = b × h** works.
- With the **diagonals**: **A = (d1 × d2) ÷ 2**. Do not forget to divide by 2.

WORKED EXAMPLE

Find the area: diagonals 12 cm and 9 cm.

Step 1 Multiply the diagonals: 12 × 9 = 108

Step 2 Divide by 2: 108 ÷ 2 = 54

Answer **54 cm²**

WORD BANK

rhombus – 4 sides, all the same length

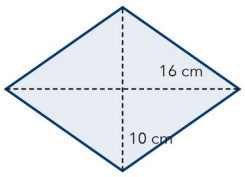
diagonal – a line from corner to opposite corner

divide by 2 – take half

PRACTICE

show your work · label units

1. Find the area of the rhombus.



Step 1 Multiply the two diagonals.

Step 2 Divide by 2.

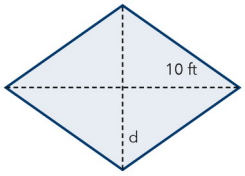
Show your work.

Answer_____ cm²

2. Find the area of each rhombus. Choose the formula that fits.

a. diagonals 14 m and 10 m _____ m² **b.** base 9 ft, height 7 ft _____ ft²

3. **Window.** A rhombus window has an area of 60 ft². One diagonal is 10 ft. Find the other diagonal *d*.



Step 1 Double the area: 2 × 60.

Step 2 Divide by the diagonal you know: ÷ 10.

Answer_____ ft

4. **Stretch.** Rhombus A has diagonals of 12 ft and 8 ft. Rhombus B has a base of 8 ft and the **same area** as A. Find the height of B.

Step 1 Find the area of A: 12 × 8 ÷ 2.

Step 2 Divide that area by 8.

Answer_____ ft

5. **Find the Error.** Ravi says a rhombus with diagonals of 18 in and 8 in has an area of 144 in². Explain his mistake and find the correct area.

Frame *Ravi forgot to ___. The area is 18 × 8 ÷ 2 = ___ in².*

_____

_____

TEST PRACTICE

choose the best answer

6. A rhombus has sides of 7 m and a perpendicular height of 5 m. What is its area?

A 12 m² **B** 17.5 m² **C** 35 m² **D** 49 m²

7. A rhombus has diagonals of 20 in and 9 in. What is its area?

Step 1 Multiply 20 × 9.

Step 2 Divide by 2.

A 29 in² **B** 45 in² **C** 90 in² **D** 180 in²

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___